

SHETH MVLU COLLEGE

SUBJECT:- WSN

Aim:Basic MANET Implementation with Packet Animation

Simulate a mobile ad-hoc network and visualize packet movement to study dynamic topology formation.

CODE:-

```
set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802_11
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ant) Antenna/OmniAntenna
set val(ifqlen) 50
set val(nn) 5
set val(rp) AODV
set val(x) 500
set val(y) 500
set val(stop) 30

set ns [new Simulator]

set tracefd [open manet.tr w]
$ns trace-all $tracefd

set namfd [open manet.nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)

set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)

create-god $val(nn)

set chan_1_ [new $val(chan)]

$ns node-config \
  -adhocRouting $val(rp) \
  -llType $val(ll) \
  -macType $val(mac) \
  -ifqType $val(ifq) \
  -ifqLen $val(ifqlen) \
  -antType $val(ant) \
  -propType $val(prop) \
  -phyType $val(netif) \
  -channel $chan_1_ \
  -topoInstance $topo \
```

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```
-agentTrace ON \  
-routerTrace ON \  
-macTrace ON \  
-movementTrace ON
```

```
for {set i 0} {$i < $val(nn)} {incr i} {  
    set node_(i)[ns node]  
}
```

```
$node_(0) set X_ 50  
$node_(0) set Y_ 50  
$node_(0) set Z_ 0
```

```
$node_(1) set X_ 150  
$node_(1) set Y_ 150  
$node_(1) set Z_ 0
```

```
$node_(2) set X_ 250  
$node_(2) set Y_ 250  
$node_(2) set Z_ 0
```

```
$node_(3) set X_ 350  
$node_(3) set Y_ 350  
$node_(3) set Z_ 0
```

```
$node_(4) set X_ 450  
$node_(4) set Y_ 450  
$node_(4) set Z_ 0
```

```
for {set i 0} {$i < $val(nn)} {incr i} {  
    node_(i) random-motion 0  
    $ns initial_node_pos node_(i) 20  
}
```

```
nsat2.0 "node_(0) setdest 400 100 15"  
nsat4.0 "node_(1) setdest 450 250 10"  
nsat6.0 "node_(2) setdest 100 400 12"
```

```
set udp0 [new Agent/UDP]  
$ns attach-agent $node_(0) $udp0
```

```
set null0 [new Agent/Null]  
$ns attach-agent $node_(4) $null0
```

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```
$ns connect $udp0 $null0
```

```
set cbr0 [new Application/Traffic/CBR]
```

```
$cbr0 set packetSize_ 512
```

```
$cbr0 set rate_ 200k
```

```
$cbr0 attach-agent $udp0
```

```
nsat5.0 "cbr0 start"
```

```
nsat25.0 "cbr0 stop"
```

```
$ns at val(stop) "ns nam-end-wireless $val(stop)"
```

```
proc finish {} {
```

```
    global ns tracefd namfd
```

```
    $ns flush-trace
```

```
    close $tracefd
```

```
    close $namfd
```

```
    puts "Simulation Completed"
```

```
    exec nam manet.nam &
```

```
    exit 0
```

```
}
```

```
$ns at $val(stop) "finish"
```

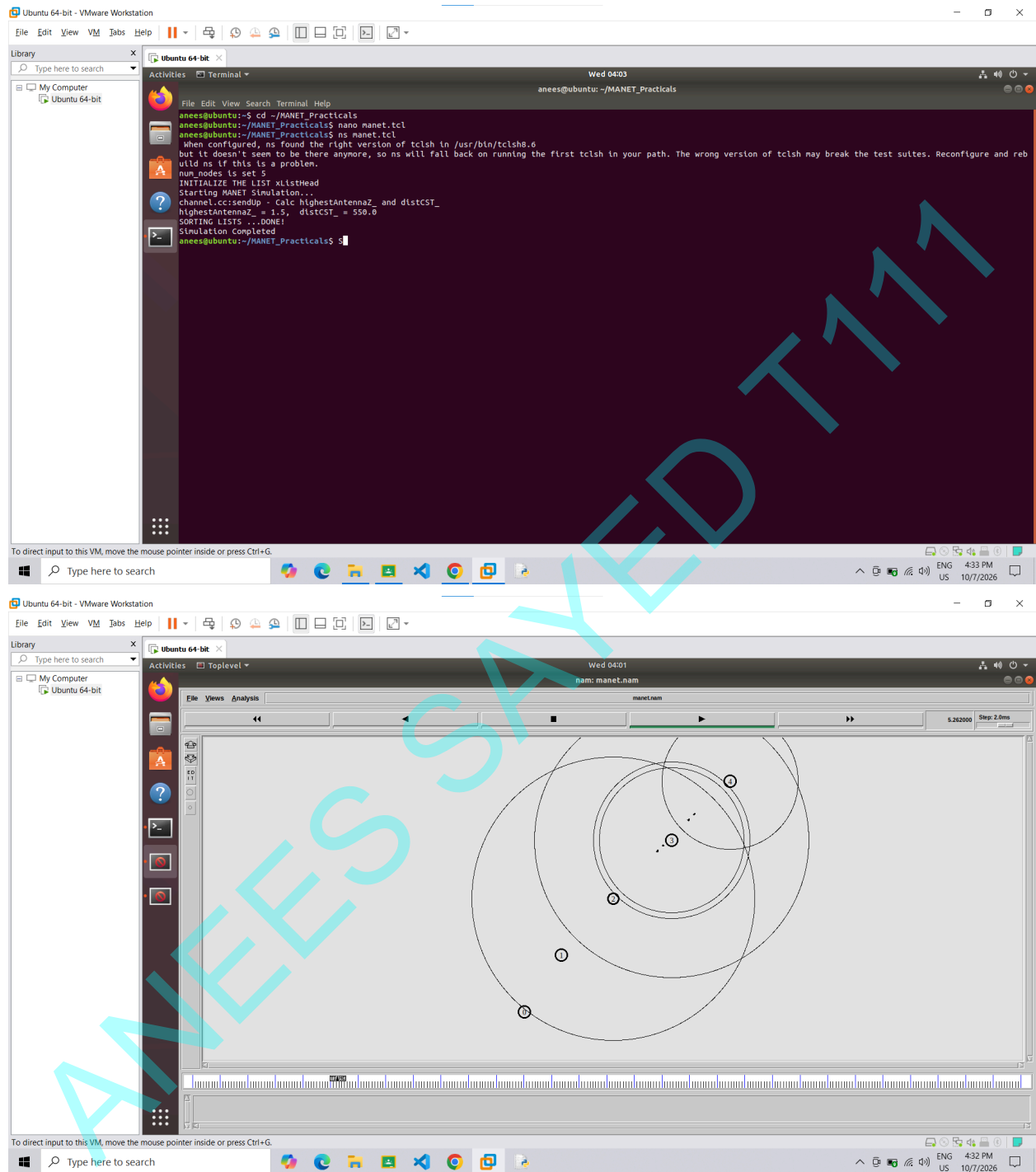
```
puts "Starting MANET Simulation..."
```

```
$ns run
```

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OUTPUT:-



Result

The MANET network was successfully simulated using NS2. Dynamic node movement and packet transmission were observed through NAM animation. The AODV routing protocol successfully established routes between mobile nodes.

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